

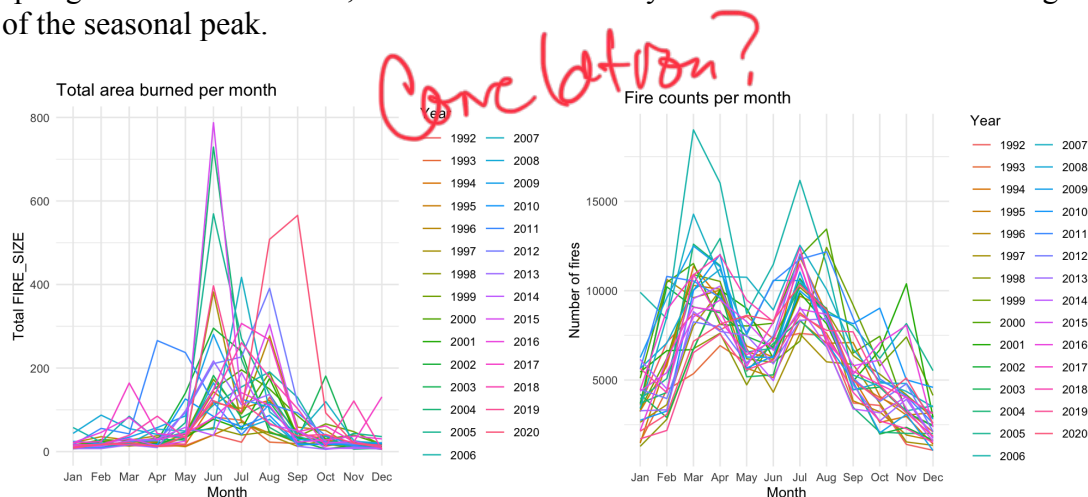
Project Proposal

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Our dataset consists of wildland fire incidents in the United States, with each row representing a single fire event. The functional variables of interest are those that vary over time or along the fire's progression: fire size (FIRE_SIZE), the discovery date (DISCOVERY_DATE, which we aggregate to months), spatial coordinates (X latitude and Y longitude), and the containment date (CONT_DATE) that marks when the fire was brought under control. These variables allow us to construct monthly trajectories of burned area and to view fire activity as a spatio-temporal process rather than a static collection of events.

The non-functional variables provide contextual information that helps explain these trajectories. Key fields include administrative location (STATE, COUNTY), cause descriptors (NWCG_CAUSE_CLASSIFICATION, NWCG_GENERAL_CAUSE), temporal grouping (FIRE_YEAR), and categorical fire size (FIRE_SIZE_CLASS). Together, these variables let us examine how patterns in fire size and timing depend on where fires occur, how they start, and whether they fall into smaller (class A–C) versus larger (class D–G) size classes, setting up more detailed functional analyses of regional and cause-specific fire regimes.

Our exploratory plots highlight several important patterns. Monthly mean fire size by year shows clear seasonality: in most years average burned area is low in winter and peaks during the late spring–summer fire season, with notable between-year differences in both the height and timing of the seasonal peak.



The companion plot of monthly fire counts reveals that the number of fires also has a strong annual cycle, but the months with the most fires are not always the months with the largest average fire size, suggesting that frequency and severity may be driven by partially different processes (for example, many small human-caused fires versus fewer large landscape-scale events).

The spatial map of ignition locations, colored from white (class A, very small fires) to dark red (class G, very large fires), shows fires concentrated in known fire-prone regions such as the

Western U.S. mountains and parts of the Southeast, and suggests geographic clusters where large fires are relatively more common.



Finally, the bar plot of NWCG_GENERAL_CAUSE indicates that a few ignition categories dominate overall fire counts, with human-related causes clearly visible alongside lightning and other natural sources, motivating questions about how cause distributions vary across space, time, and fire-size classes.

- Handwritten note: Choose either question 1 or 2 but not both*
1. How has the seasonal trajectory of total burned area changed across years, and are there detectable long-term trends in fire severity?

Monthly total burned area, aggregated from individual fire size records, constitutes the primary functional response of interest. The covariate of interest here is FIRE_YEAR, which allows each year's twelve-month burned-area profile to be treated as a single functional observation and compared across time. We can then ask whether the shape of the seasonal curve (its peak timing, amplitude, and duration) has shifted systematically over the decades covered by the dataset. This question is scientifically important because changes in the fire season's intensity and length are among the most direct ecological signatures of climate change. A fire manager or land agency scientist would want to know not only whether fires are getting larger on average, but whether the window of high risk is expanding, since that information directly informs staffing, resource pre-positioning, and prescribed burn scheduling.

- Handwritten note: Stationary time series?*
- Handwritten note: implement a change point model*
2. Does the monthly frequency of fire ignitions follow the same seasonal pattern as burned area, or do fire count and fire severity exhibit distinct temporal dynamics?

While Question 1 focuses on area burned, this question treats the monthly count of fire ignitions as a separate functional variable, again with FIRE_YEAR as the primary covariate so that annual ignition-frequency curves can be compared to annual burned-area curves. The exploratory plots

~~W/ aforementioned EDA~~

FPCA between the two curves & fit a ^{non-linear} regression model between the two

already hint that frequency and severity are not perfectly synchronized: some months accumulate many small fires while others see fewer but far larger events. Formally characterizing this divergence and testing whether it varies by year would help distinguish between regimes dominated by high-frequency, low-severity ignitions (often associated with human activity) and those characterized by infrequent but catastrophic fires (more commonly driven by climate and fuel accumulation). For the scientist who collected this data, resolving this distinction has direct implications for cause-attribution modeling and risk forecasting.

5 3. Does the seasonal profile of burned area differ across geographic regions or states, and can spatially distinct fire regimes be identified from the functional data?

This question introduces STATE (or a broader regional grouping) as the key non-functional covariate, and asks whether the monthly burned-area trajectories differ meaningfully across space. The spatial map of ignition locations already suggests geographic clustering of large fires in the Western U.S. and parts of the Southeast, but a functional analysis would go further by comparing the full seasonal curve, not just summary statistics, across regions. Identifying spatially distinct fire regimes matters for resource allocation: a federal land management agency needs to know whether the peak fire season in the Southwest precedes or overlaps with that of the Pacific Northwest, since simultaneous demand for suppression resources across regions creates logistical bottlenecks. State-level functional comparisons could also reveal whether certain jurisdictions have experienced disproportionate increases in large-fire activity over time.

4 4. Does the cause of ignition, whether natural or human-related, systematically alter the distribution or seasonal trajectory of fire size, and has this relationship changed over time?

The covariates in this question are NWCG_GENERAL_CAUSE, as a categorical grouping variable, and FIRE_YEAR, which allows us to assess whether any cause-related differences in fire size or seasonality have strengthened or weakened over time. Together these covariates let us ask whether fires of different origins differ not just in their mean size but in their entire distributional or temporal profile. Human-caused fires may be more uniformly distributed across the calendar year, while lightning-ignited fires may show a sharper seasonal peak tied to monsoonal storm patterns. This question is of direct interest to fire prevention programs, which must decide how to allocate public education and enforcement resources relative to suppression investments, and becomes especially pressing as wildland-urban interface expansion increases human ignition pressure in fire-prone landscapes.

2 X 5. How does time-to-containment vary as a function of fire size class, geographic location, and year, and is there evidence of systematic improvement in suppression efficiency over time?

where's the function?

The covariates here are FIRE_SIZE_CLASS and STATE as non-functional grouping variables, and FIRE_YEAR as a temporal covariate, allowing us to examine whether containment duration,

computed as the difference between containment day-of-year and discovery day-of-year, differs across size categories, varies by state, and has trended in any consistent direction over time. For the scientist or agency administrator who assembled this dataset, this question has direct operational relevance: if containment times for large fires (class E through G) have not improved despite advances in aerial suppression and early detection technology, that is an important finding that would motivate deeper investigation into resource adequacy and climate-driven fire behavior.